

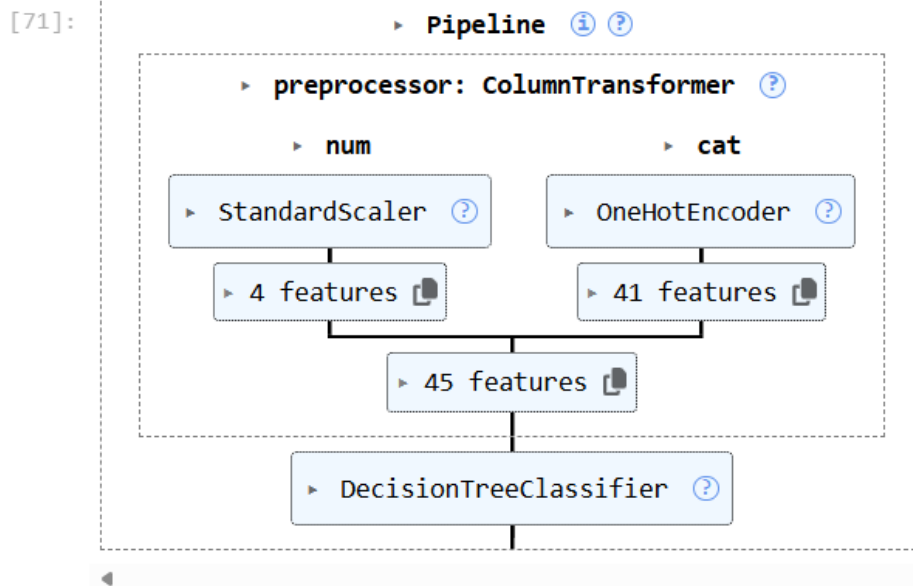
Decision Tree

```
[68]: from sklearn.tree import DecisionTreeClassifier
```

```
[69]: tree_model = Pipeline([
    ("preprocessor", preprocessor),
    ("classifier", DecisionTreeClassifier(
        random_state=42
    ))
])
```

```
[70]: # Train:
```

```
[71]: tree_model.fit(X_train, y_train)
```



```
[72]: # Then predict:
```

```
[73]: y_pred_tree = tree_model.predict(X_test)
```

```
[74]: # get probabilities for ROC-AUC:
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```
[75]: y_proba_tree = tree_model.predict_proba(X_test)[:, 1]
```

```
[76]: # evaluate it
```

```
[77]: from sklearn.metrics import accuracy_score, classification_report, roc_auc_score

print("Accuracy:", accuracy_score(y_test, y_pred_tree))

print("\nClassification Report:")
print(classification_report(y_test, y_pred_tree))

print("ROC-AUC:", roc_auc_score(
    y_test,
    y_proba_tree
))
```

Accuracy: 0.7210787792760823

Classification Report:

	precision	recall	f1-score	support
No	0.81	0.80	0.81	1035
Yes	0.48	0.49	0.48	374
accuracy			0.72	1409
macro avg	0.64	0.65	0.65	1409
weighted avg	0.72	0.72	0.72	1409

ROC-AUC: 0.6476762510010592